

Diana Spahieva

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WhoAmI

My main interests lie in the field of Data Science with an emphasis on applied machine learning and software development, as well as data engineering. I aspire to be involved in a company that aims to change the world for the better, especially in terms of sustainability.

TECHNICAL SKILLS

Proficiency: Advanced Intermediate Fundamental Learning

Programming Languages: Python R SQL MATLAB Bash JavaScript

Deep Learning Frameworks: Keras TensorFlow PyTorch

Data Engineering Tools: AWS Google Cloud Platform (GCP) Apache Spark ETL (Extract, Transform, Load) CICD Jenkin
DevOps Virtual Machines Databases (SQLite MongoDB PostgreSQL) Docker Kubernetes

Data Science Tools: Pandas NumPy Scikit-learn Matplotlib Seaborn OpenCV Natural Language Processing: NLP (NLTK SpaCy Hugging Face) Power BI Statsmodels Social Network Analysis (SNA) Explainable AI Tableau

Libraries & Tools: Linux VSCode Git Agile Version Control Jira Trello Confluence Selenium Web Scraping
Django

Languages: English Bulgarian Dutch German

EXPERIENCE

Freelancing

upwork

July 2024 – Present

NL

- Created data-driven dashboards using **Django**, **PostgreSQL**, and **JavaScript**, offering clients insightful visualizations and analytics tailored to their business needs.
- Designed and implemented database schemas, optimized queries, and ensured the security of user data through best practices in database management and encryption.
- Collaborated with clients to gather requirements and provide solutions that were cost-effective and aligned with business goals, delivering projects on time.

Data Engineer Intern

NXP Semiconductors

August 2023 – June 2024

Eindhoven, NL

- Designed a monitoring system in **Splunk** to detect data gaps in an R&D **data lake**.
- Developed a **Splunk dashboard** to provide real-time monitoring and visibility into data streams and alert the responsible team when gaps or inconsistencies were detected.
- Utilized **AWS Glue jobs** and **Lambda functions** for **data orchestration**, integrating on-premise databases with the serverless **AWS** architecture of the data lake.
- Tested new features locally using **Docker** containers to validate functionality and dependencies before deploying to the **AWS** environment for further integration testing.
- Enhanced the **scalability** and **performance** of the data lake by addressing potential bottlenecks and recommending process optimizations for **long-term reliability**.
- Tracked development tasks using **Jira** and documented the end-to-end workflow in **Confluence** to enable knowledge transfer and future improvements by the data lake management team.

PROJECTS

Identifying high potential customers to drive sales in consultancy

Feb 2023 – June 2023

Data Science Project

Python, Web scraping, Selenium, Power BI, NLP

- Developed a **data scraping** solution to extract company insights from LinkedIn and Nationale Vacaturebank, including metrics such as employee counts and job vacancies.
- Developed a **Power BI dashboard** that enabled the sales team to identify high-potential new customers through advanced filtering and sorting options.
- Created a customer segmentation model based on company size, growth phase, and job posting activity to prioritize outreach efforts.

Biodiversity Insights Network – Automated Data Pipeline for Strategic Collaboration

Aug 2022 – Dec 2022

Data Engineering Project

Python, NLP, Power BI, SNA

- Developed an end-to-end automated **data pipeline** to process biodiversity documents, applying natural language processing (NLP) for topic extraction, actor identification, and network generation.
- Implemented **topic modeling** using **BERT** to uncover key themes and topics from documents, facilitating the identification of connections between individuals and organizations within the biodiversity domain.
- Built a **bipartite network** to visualize relationships between actors (people/organizations) and topics, which was further projected into a direct actor-to-actor network to reveal critical collaboration patterns. Visualized in Power BI dashboard.

BDVT - AN INTERACTIVE TOOL TO EXPLORE AND ANALYZE THE BHIC DATA

Feb 2021 – June 2021

Front and back end development

Python, Django, GUI design

- Developed the **BHIC Data Visualization Tool (BDVT)** for visualizing genealogical data from the Brabant Historical Information Center (BHIC) dataset. The tool was built with a **Python** back-end using the **Django** web framework, and deployed on a **TU/e-provided server**.
- It enables interactive exploration of historical data of Brabant's inhabitants through diverse visualizations such as **line plots**, **word clouds**, **pedigree diagrams**, **population pyramids**, **network graphs**, **treemaps**, and **line charts**.

EDUCATION

Tilburg University, Eindhoven University of Technology

's-Hertogenbosch, The Netherlands

M.S. Data Science in Business and Entrepreneurship

Sep 2022 – June 2024

• **Modules include:** Data Intrapreneurship in Action; Data Mining; Data Engineering; Strategy and Business Models; Social Network Analysis for Data Scientists; Data Consultancy in Action; Interactive and Explainable AI Design; Intellectual Property and Privacy; Deep Learning; Natural Language Processing

Tilburg University, Eindhoven University of Technology

Eindhoven, The Netherlands

B.S. in Data Science

Sep 2018 – June 2022

• **Modules include:** Programming (Python); Perspectives on data science; Data Statistics (R); Understanding the information Society; Creative thinking (Python); Data mining (WEKA); Data Analytics for engineers (Python); Statistical Computing (R); Data science ethics; DBL Data challenge (Python); Law and Data Science; Engineering design; Data challenge 1 to 3 (Python); Visualization (Python); Process mining (Python); Minor in Data Science and Entrepreneurship

INTERESTS

- **Sport:** I am a very active person and sports play an important role in my lifestyle. In my free time I enjoy yoga, running, fitness and hiking.
- **Travel & Music:** Exploring new places and immersing myself in different cultures are among my greatest passions. I particularly enjoy combining travel with attending live music events, discovering new destinations during the day and concerts in the evening.
- **Board Games & Storytelling:** I enjoy playing Dungeons & Dragons and other board games, appreciating the mix of creativity, strategy, and shared storytelling they bring to social gatherings.